```
from google.colab import files
uploaded = files.upload()
```

Choose Files Screenshot… 102039.png

Screenshot 2026-07-22 102039.png(image/png) - 609392 bytes, last modified: 7/22/2026 - 100% done

```
Saving Screenshot 2026-07-22 102039.png to Screenshot 2026-07-22 102039.png
```

```
import cv2
import matplotlib.pyplot as plt

# Get the name of the uploaded file
image_filename = list(uploaded.keys())[0]

# Load the image
image = cv2.imread(image_filename)

if image is None:
    raise FileNotFoundError(f"Image '{image_filename}' not found. Please ensure it was uploaded correct

image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# ------------------------------------------
# Create an aliased image (without filtering)
# ------------------------------------------
aliased = cv2.resize(
    image,
    (image.shape[1] // 4, image.shape[0] // 4),
    interpolation=cv2.INTER_NEAREST
)

# ------------------------------------------
# Corrective approach: Anti-aliasing
# Apply Gaussian Blur before downsampling
# ------------------------------------------
```

```
blurred = cv2.GaussianBlur(image, (5, 5), 1.0)

anti_aliased = cv2.resize(
    blurred,
    (image.shape[1] // 4, image.shape[0] // 4),
    interpolation=cv2.INTER_AREA
)

# ----------------------------------------
# Display Results
# ----------------------------------------
plt.figure(figsize=(12, 4))

plt.subplot(1, 3, 1)
plt.imshow(image)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(aliased)
plt.title("Aliased Image")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(anti_aliased)
plt.title("Anti-Aliased Image")
plt.axis("off")

plt.tight_layout()
plt.show()
```


Original Image
Aliased Image
Anti-Aliased Image